CALCULUS · UNIT 4: APPLICATIONS OF DERIVATIVES

Package Design Studio

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students design packaging for a juice company. Box and can dimensions are optimized, a cone-shaped filling tank requires related rates, and a sales curve is analyzed with first and second derivatives and the MVT/EVT.

Comparing the math-optimal can to real cans opens a discussion of modeling assumptions.

Standards Addressed

- AP FUN-4 Analytical applications: extrema, MVT, EVT, concavity
- AP CHA-3 Related rates
- AP FUN-4.C Optimization
- AP CHA-3.F Linearization

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a soda can and a cereal box. Part 1 optimization.
Day 2	Part 2 related rates with the cone tank.
Day 3	Part 3 sales curve and proposal.

Materials

- Student handout and Design Data Cards
- Graphing calculators
- Paper, scissors, and tape to build the open box
- A real can and ruler

How the Three Versions Differ

Level	What changes
CP2	Open-box maximum, cylinder related rate, and increasing/decreasing sales.
CP1	Can surface area optimization, cone related rates, concavity and inflection, and the MVT.
Honors	Cost-weighted optimization, leaking tank model, linearization, EVT, and savings analysis.

Watch For (Common Misconceptions)

- Forgetting to eliminate a variable before differentiating.
- Substituting values before differentiating in related rates.
- Not checking endpoints for absolute extrema.

Extension Ideas

- Build and measure the optimal box.
- Research why cans are shaped the way they are.
- Optimize a cereal box for a fixed volume.

Answer Key – CP2

Part 1: Open Box

#	Task	Answer
1	Write the volume $V(x) = x(30 - 2x)(20 - 2x)$.	$V(x) = 4x^3 - 100x^2 + 600x$
2	What is the domain of x?	$0 < x < 10$
3	Find $V'(x)$ and the critical point in the domain.	$V' = 12x^2 - 200x + 600 = 0 \rightarrow x \approx 3.92$ cm
4	Find the maximum volume.	$V(3.92) \approx 1{,}056$ cm^3

Part 2: Filling the Tank

#	Task	Answer
5	A cylinder tank has radius 2 m, so $V = 4\pi h$. Find $\frac{dh}{dt}$.	$\frac{dV}{dt} = 4\pi\frac{dh}{dt} \rightarrow \frac{dh}{dt} = \frac{2}{4\pi} \approx 0.159$ m/min
6	Explain why $\frac{dh}{dt}$ is constant for a cylinder.	The cross-section area does not change with height.
7	A cone tank fills faster at the bottom. Explain why using its shape.	The cross-section is smaller near the bottom, so the same volume raises the level more.
8	If juice drains at 0.5 m^3/min while filling at 2 m^3/min, what is the net $\frac{dV}{dt}$?	1.5 m^3/min

Part 3: Sales Curve

#	Task	Answer
9	Find $S'(t)$ and when sales are increasing on $0 \le t \le 12$.	$S' = -3t^2 + 24t + 60 = -3(t - 10)(t + 2) \rightarrow$ increasing for $0 < t < 10$
10	When are total sales at a maximum? Find the value.	$t = 10$; $S(10) = 800$ thousand cans
11	Write one recommendation for the company using your results.	Answers vary; e.g., launch a new flavor before month 10.

Reflection

#	Prompt	Look for
R1	Why do you check the endpoints and critical points?	The maximum or minimum on a closed interval happens at one of them.
R2	What does $S'(t) = 0$ mean for sales?	Sales stop increasing and start decreasing (a turning point).

Answer Key – CP1

Part 1: Optimize the Can

#	Task	Answer
1	Write h in terms of r and A as a function of r alone.	$h = \frac{355}{\pi r^2}$; $A(r) = 2\pi r^2 + \frac{710}{r}$
2	Find A′(r) and the critical point.	$A' = 4\pi r - \frac{710}{r^2} = 0 \rightarrow r = \sqrt[3]{(\frac{355}{2\pi})} \approx 3.84$ cm
3	Find h and compare h with the diameter. Justify that this is a minimum.	h ≈ 7.67 cm ≈ 2r; A″ > 0 → minimum
4	Real cans have r ≈ 3.3 cm and h ≈ 10.4 cm. How much more aluminum do they use (in cm²)?	A(3.3) ≈ 283.6 vs. A(3.84) ≈ 277.5 → about 6 cm² more

Part 2: Cone Tank

#	Task	Answer
5	Use similar triangles to write r in terms of h and V in terms of h.	$r = \frac{h}{3}$; $V = \frac{\pi h^3}{27}$
6	Find $\frac{dh}{dt}$ when h = 3 m.	$2 = \frac{\pi h^2}{9}\frac{dh}{dt} \rightarrow \frac{dh}{dt} = \frac{18}{9\pi} \approx 0.637$ m/min
7	Find how fast the radius of the juice surface is increasing at h = 3.	$\frac{dr}{dt} = \frac{1}{3}\frac{dh}{dt} \approx 0.212$ m/min
8	Find how fast the surface area of the juice top is increasing at h = 3.	$A = \pi r^2$, $\frac{dA}{dt} = 2\pi r\frac{dr}{dt} = 2\pi(1)(0.212) \approx 1.33$ m²/min

Part 3: Sales Analysis

#	Task	Answer
9	Find intervals of increase/decrease and concavity, and the inflection point.	Increasing (0, 10), decreasing (10, 12); S″ = −6t + 24 → concave up (0, 4), down (4, 12); inflection t = 4
10	Interpret the inflection point for the marketing team.	At month 4, sales are growing fastest (S′ is maximum: 108 thousand/month).
11	Use the MVT on [0, 10] to show that sales grew at exactly 80 thousand cans/month at some time. Find it.	Average rate $\frac{800}{10} = 80$; $-3t^2 + 24t + 60 = 80 \rightarrow t \approx 0.95$ or 7.05

Reflection

#	Prompt	Look for
R1	Why do real cans not use the optimal shape?	Other factors: grip, stacking, labels, and the thick top and bottom.
R2	What is the difference between where S is maximum and where S′ is maximum?	S is maximum when sales stop growing; S′ is maximum when sales grow fastest (inflection).

Answer Key – Honors

Part 1: Cost Optimization

#	Task	Answer
1	Write the cost C(r).	$C = 0.05(2\pi r^2) + 0.02(2\pi r)\left(\frac{355}{\pi r^2}\right) = 0.1\pi r^2 + \frac{14.2}{r}$
2	Find the radius that minimizes cost and the height.	$C' = 0.2\pi r - \frac{14.2}{r^2} = 0 \rightarrow r = \sqrt[3]{\left(\frac{71}{\pi}\right)} \approx 2.83$ cm; h ≈ 14.1 cm
3	Compare with the minimum-area can. Why is the cost-optimal can taller?	Expensive ends are made smaller, so the can becomes taller and thinner.
4	Find the minimum cost per can and the savings per million cans versus r = 3.84 cm.	C(2.83) ≈ 7.53¢; C(3.84) ≈ 8.33¢; savings ≈ 0.8¢ × 1,000,000 ≈ $8,000

Part 2: Tank Dynamics

#	Task	Answer
5	Write $\frac{dV}{dt}$ and then $\frac{dh}{dt}$ as a function of h.	$\frac{dV}{dt} = 2 - 0.1h$; $\frac{dh}{dt} = \frac{9(2-0.1h)}{\pi h^2}$
6	Find $\frac{dh}{dt}$ when h = 3 m.	$\frac{9(1.7)}{9\pi} \approx 0.541$ m/min
7	Is there a height where the level stops rising? Is it inside the tank?	2 − 0.1h = 0 → h = 20 m; no, the tank (6 m) fills first.
8	Use linearization of $V(h) = \frac{\pi h^3}{27}$ at h = 3 to estimate the volume added when h goes from 3 to 3.1 m.	$dV \approx V'(3)(0.1) = \frac{\pi(9)}{9}(0.1) \approx 0.314$ m³

Part 3: Sales Theorems

#	Task	Answer
9	Use the EVT to find the absolute max and min of S on [0, 12].	Candidates: S(0) = 0, S(10) = 800, S(12) = 720 → max 800, min 0
10	Find the maximum of the growth rate S′ on [0, 12] and justify it.	S″ = 0 at t = 4; S′(4) = 108 (S‴ < 0) → max growth 108 thousand/month
11	Write the design proposal: can dimensions, tank controls, and the sales timeline, each justified with calculus.	Answers vary.

Reflection

#	Prompt	Look for
R1	How did changing the cost structure change the optimal shape?	Making the ends expensive pushed the design toward smaller ends and a taller can.
R2	Why does linearization work well for small changes?	Near the point the curve is close to its tangent line.